ARAZVERA TECHNOLOGIES | ENGINEERING PORTFOLIO | 10 NOV 2026

LEVEL 4

Enterprise Resilience & Security Engineering

Acceptance-tested backend reliability, secure service recovery and safe change engineering

SPECIALIST CAPABILITY CATALOG — VERSION 0.3.2

This document presents a real, **isolated single-host engineering laboratory**. It is not a customer production deployment, Enterprise security certification or uptime guarantee.

Engineering proposition

Complex platforms often fail across the boundaries between HTTP APIs, PostgreSQL transactions, tenant identity, background workers, monitoring and release control. ARAZVERA Level 4 builds reproducible, bounded evidence and reviewable corrective engineering within separately authorized environments.

Six specialist service families

Engineering service	Representative scope
Architecture & Reliability Audit	Service dependency mapping, incident pathways, operational risk, SLO gaps and acceptance design
Multi-Tenant Security Hardening	RS256/JWKS validation, PostgreSQL FORCE RLS, restricted roles, tenant-scope and replay negative tests
Transactional Recovery Engineering	Durable outbox, idempotency, fenced leases, bounded retries, dead letters, Worker recovery and logical restore
Safe Staging Release Design	Stable/Candidate HTTP routing, 200-request measured Canary, HOLD decisions and authorized lab-only rollback
Incident & Operations Readiness	Real service probes, persisted incident history, queue and latency metrics, evidence and explicit unknowns
Backend Reliability Remediation	Bounded FastAPI/Python/PostgreSQL/Docker integration work with regression tests and customer handover

Lab evidence demonstrates engineering capability, not a customer's deployed performance.

Verified technical evidence

Every result below reflects an **executed** test. Individual test groups overlap and must not be added together.

Verification	Actual result and scope
Latest private GitHub Actions	**4/4 jobs passed** on commit **7dcea759**, run **42787753161**; Python 3.11/3.12, PostgreSQL 16 and disposable multi-service Docker
Full local Docker Pytest	**98 passed**, zero failures; 2 non-blocking cache warnings
Local Python unit suite	**76 passed**, zero failures
Real Chrome/Selenium UI	**13/13 checks passed**: probes, release HOLD, Canary/rollback, evidence and session disconnect
Real synthetic Canary acceptance	**3/3**: 200 candidate HTTP observations, a passing candidate route and failing HTTP 500 HOLD/Stable recovery
Scoped API and PostgreSQL outage	**17/17**: labelled container stop/start, non-healthy readiness, persisted data and restoration
PostgreSQL logical restore	**8/8** checks in a second disposable PostgreSQL container and volume
Independent Worker restart	**7/7** with durable event receipt after recovery
Docker restart policy	Nine long-running containers inspected as **unless-stopped**; bootstrap, restore and test containers remain **restart: no**

Laboratory architecture

The isolated Compose project contains a FastAPI Control Plane and English dashboard, PostgreSQL 16, independent Worker, observer, synthetic downstream, ephemeral RS256 identity provider and separate Stable/Candidate backend containers. The only host-published endpoint is the operator API on Windows loopback; the data and identity backends are internal.

The Canary engine sends 200 actual candidate HTTP requests, stores measured latency/error evidence and changes only the synthetic staging route. A faulty candidate results in a durable HOLD and route rollback. A manual rollback is independently recorded.

The updated Docker **unless-stopped** policy allows its nine services to restart when a functioning Docker Engine resumes. **Windows host and Docker daemon reboot were not exercised**, and their automatic startup is not guaranteed.

Engagement design, deliverables and exclusions

Acceptance-based engineering process

Stage	Deliverable
1. Assessment	Written authorization boundary, reproducible failure, risk register and acceptance requirements
2. Reproduction	Isolated baseline, negative conditions and measured failure evidence
3. Remediation	Reviewed backend, identity, recovery or staging release changes
4. Verification	Real regression tests, bounded fault/recovery exercise, results and documented limitations
5. Handover	Code review package, English runbook, release/rollback rules and acceptance register

Commercial terms are proposal-specific. Hosting, credentials and external API services are client-owned and require explicit authorization. No delivery price, commercial outcome, certification or existing Enterprise customer relationship is guaranteed.

What these tests do not prove

- No live customer production deployment, independent security certificate, guaranteed uptime or customer SLO.
- The synthetic RS256 identity and PostgreSQL FORCE RLS tests do not prove enterprise customer SSO or corporate PKI integration.
- A single-host logical restore and scoped container restart do not prove multi-site disaster recovery, offsite RPO/RTO or host failover.
- Lab Canary and rollback do **not** deploy or revert a customer's application image.
- Passing basic secret scans and CI is not an independent penetration test, complete dependency audit or signed software provenance.
- SHA-256 evidence digests provide content checksums, not cryptographic signer identity.
- No confidential source, credentials, customer information, unrelated systems or production trading resources are published.

Public resources

ARAZVERA TECHNOLOGIES
Company GitHub
Engineering Portfolio
Level 4 Service Summary

This is an evidence-backed capability statement. All production work requires separate customer authorization and acceptance.